



Debre Berhan university

College of Computing

Department : Software Engineering

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Intelligence

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Group Assignment

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Title

Smartwatch Step Counter AI Agent

AI-based model-driven system that tracks user steps, estimates distance, and learns walking behavior automatically.

Problem Statement

Key idea: Tracking physical activity manually is difficult and inaccurate.

- People want to monitor daily activity
- Manual counting of steps is not practical
- Distance estimation without tools is inaccurate
- Different users have different walking styles

So we need a system that automatically tracks steps and estimates distance in real time.

Project Goal

Main Goal:

Build a **Model-Based AI Agent** that tracks steps and estimates distance accurately.

Agent Responsibilities:

- Detect user movement
- Count steps automatically
- Estimate distance using stride length
- Adjust based on activity (walking, running)
- Improve accuracy using GPS calibration
- Display results in real time
- Store activity history

Why This Is a Model-Based AI Agent

Model-Based Agent = Uses memory + internal model

State:

- Number of steps
- Distance traveled
- Activity type
- Stride length
- Confidence level

Model:

- StrideModel (learns user walking behavior)

Actions:

- Count steps
- Estimate distance
- Update stride using GPS
- Display results

The agent **remembers past data and improves over time.**

Autonomous Behavior (Agent Loop)

Flow:

1. Sensor reads movement
2. Agent detects step
3. Steps are updated
4. Distance is calculated
5. GPS calibrates stride
6. Results displayed
7. Loop continues automatically

Works continuously without manual input.

Walking Scenario

Agent Behavior:

- Detects regular movement
- Counts steps steadily
- Uses normal stride length
- Distance increases gradually

Example output:

Estimating 150 m

Running Scenario

Agent Behavior:

- Detects higher acceleration
- Applies larger stride factor
- Distance increases faster

Example:

- Running stride > walking stride

More realistic estimation.

GPS Calibration (Learning)

Important AI Feature

- System receives GPS distance
- Calculates real stride:

Stride = GPS distance / Steps

- Updates internal model
- Improves confidence

Example output:

GPS calibration -> stride 0.84 m

This shows **learning capability**.

System Architecture

Main Components:

1. Sensors

- MotionSensor
- Reads acceleration (ax, ay, az)

2. Agent

- SmartwatchAgent
- Makes decisions

3. Model

- StrideModel
- Predicts distance

4. Environment

- UserMovementSimulator
- Simulates user activity

Backend Modules Overview

- **Motion Sensor** → detects movement
- **Stride Model** → estimates distance
- **Agent** → decision making
- **Simulator** → generates user activity
- **Runner** → controls simulation loop

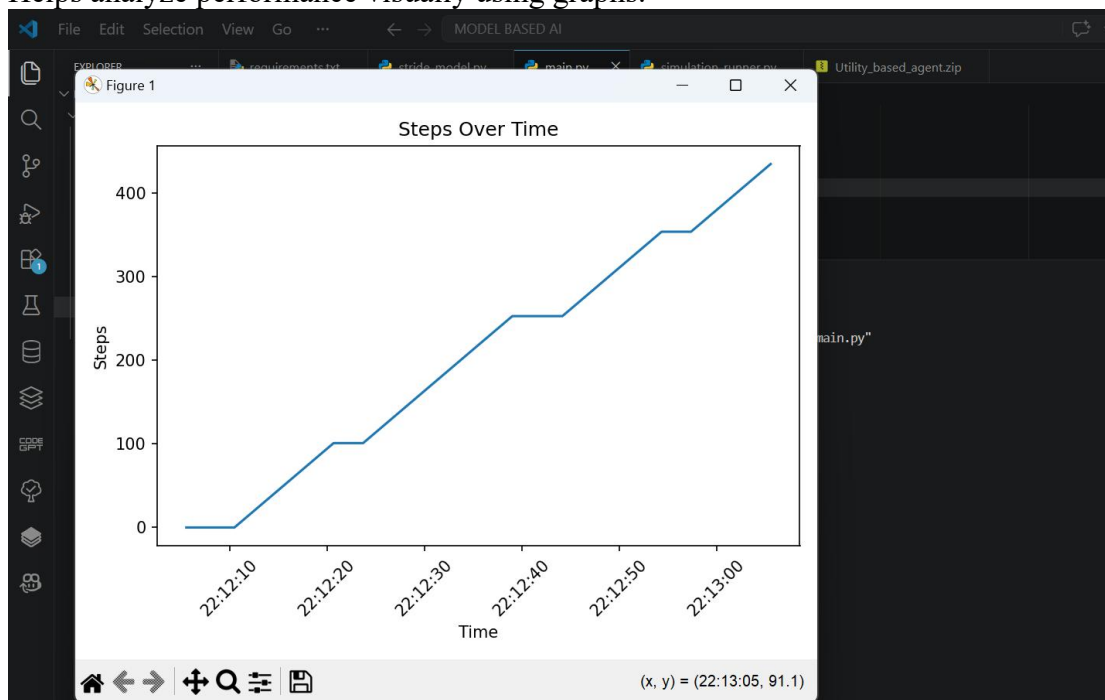
Shows good modular design.

Visualization (Graphs)

System displays:

- Steps over time
- Distance over time

Helps analyze performance visually using graphs.



Autonomous Features

System is fully automatic:

- No manual step input
- Continuous simulation loop
- Automatic step detection
- Automatic distance calculation
- Automatic GPS calibration

Works independently (no human control needed)

Logging & History

System stores:

- Time
- Steps
- Distance

Enables tracking and analysis.

Technologies Used

- Python
- NumPy (calculations)
- Matplotlib (graphs)

Conclusion

Key Achievements:

- Built a **Model-Based AI Agent**
- Implemented **step detection system**
- Created **distance estimation model**
- Added **learning using GPS calibration**
- Visualized results using graphs

This project demonstrates how AI agents can be used in **real-world wearable devices**.

FINAL PRESENTATION LINE

“This project demonstrates a model-based AI agent that senses movement, learns user behavior, and continuously improves its accuracy in estimating distance, similar to real smartwatches.”

The out put is like here:

